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ResumeHub - An AI & Blockchain Powered Platform for Hiring

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Table of Contents

0. Forenote and Guide	3
1. Executive Summary	4
2. Introduction	5
2.1 Resume Fraud and Misinformation	5
2.2 Current Issues with Job Hiring Process	5
2.2.1 Inefficiency in Background Verification Checks (P1)	6
2.2.2 Inefficiency and Cost of Fragmented Use of Hiring Softwares (P2)	6
2.2.3 Data Vulnerability in Storage and Transit (P3)	6
2.3 Formulating Opportunity Statements	7
3. Market and Competitor Analysis	8
3.1 Market Analysis	8
3.2 Competitor Analysis	9
4. Proposed Solution	10
4.1 Core User Features	10
4.1.1 Students	11
4.1.2 Companies	11
4.2 Core Technologies	12
4.2.1 Vector Embeddings	12
4.2.2 GenAI and RAG	13
4.2.3 Blockchain and Web3	14
5. Technical Architecture	15
6. Value Proposition and Core Principles	17
6.1 Value Proposition	17
6.2 Business Value	18
7. Feasibility Analysis	19
7.1 Financial	19
7.2 Technical	19
7.3 Regulatory	20
8. Conclusion	21
9. References	22

0. Forenote and Guide

To support your reading of this report, we recommend referring to the following supplementary documents included in the project zip folder:

1. **Folder Structure and Functionality** - Refer to this document at any time to understand more about the code folder structure. All of the code can be found under the “ResumeHub” folder.
2. **User Journey and Guide** – Refer to this alongside Sections 4.1.1 and 4.1.2 to view ResumeHub’s features from the perspectives of both job seekers and employers.
3. **App Architecture** – Refer to this alongside Section 5 for a deeper understanding of the architectural decisions behind ResumeHub.
4. **Setup and Deployment Guide** – Refer to this after Section 5 for step-by-step instructions on how to deploy and run ResumeHub locally or online.

1. Executive Summary

In an increasingly digital and competitive job market, both employers and students face rising challenges in the hiring process, ranging from resume fraud and inefficient verification to fragmented platforms and growing concerns over data privacy.

In this report, we will introduce ResumeHub, an end-to-end solution designed to bridge the gaps in hiring processes. ResumeHub is an AI-powered, blockchain-integrated hiring platform designed to streamline credential verification, unify hiring workflows, and enhance data security.

The report begins by outlining three core pain points in the hiring ecosystem, followed by opportunity statements that guide our problem-solution fit. We then evaluate existing market solutions and highlight the gaps they fail to address, establishing the need for a new approach.

Next, we introduce ResumeHub's key features and technical architecture, built on a hybrid Web2 – Web3 model using technologies such as FastAPI, React, Supabase, PostgreSQL, and the Polygon blockchain. We describe how core components and key learnings from classroom like vector embeddings, Retrieval-Augmented Generation (RAG), and Soulbound Tokens contribute to our goals of security, automation, and verifiability.

Finally, we conduct a feasibility analysis, covering technical, regulatory, financial, and adoption considerations which is supported by current market trends and government initiatives like Singapore's Career & Skills Passport. We conclude with key recommendations and future development plans to scale ResumeHub into a viable solution for next-generation hiring.

2. Introduction

2.1 Resume Fraud and Misinformation

In today's competitive society, resume fraud has become a widespread issue, with research indicating that up to 70% of job applicants exaggerate or fabricate their qualifications (Robinson, 2023). The increasing availability of counterfeit degrees online further exacerbates this issue, enabling unqualified individuals to misrepresent their credentials with ease (Tang, 2023).

Such fraudulent activities undermine the integrity of the hiring process and create unfair competition among job seekers. Employers, faced with an overwhelming number of applications, may struggle to distinguish between legitimate and falsified qualifications, ultimately making poor hiring decisions.

The inability to effectively verify candidate credentials has led to a growing trust deficit in hiring. Recruiters frequently encounter difficulties in differentiating between genuine and fraudulent applications, which can result in false positives, where unqualified individuals are hired, or false negatives, where competent candidates are rejected due to verification challenges. This misjudgment negatively affects both job seekers and employers, leading to increased employee turnover and decreased workplace efficiency. Additionally, organisations that unknowingly hire individuals with falsified credentials may suffer reputational damage, legal liabilities, and compromised workplace safety (Schooley, 2024). Addressing these trust issues is critical for ensuring a transparent and reliable hiring process that benefits all stakeholders involved.

2.2 Current Issues with Job Hiring Process

To tackle the issue of resume frauds today, HR would conduct background verification checks on shortlisted candidates prior to their onboarding using platforms provided by companies such as Sterling Risk and HireRight (Mujibah, 2024).

However, our team's analysis of the hiring process reveals certain flaws which are briefly illustrated in Figure 2.2 and discussed in detail in the subsequent subsections.

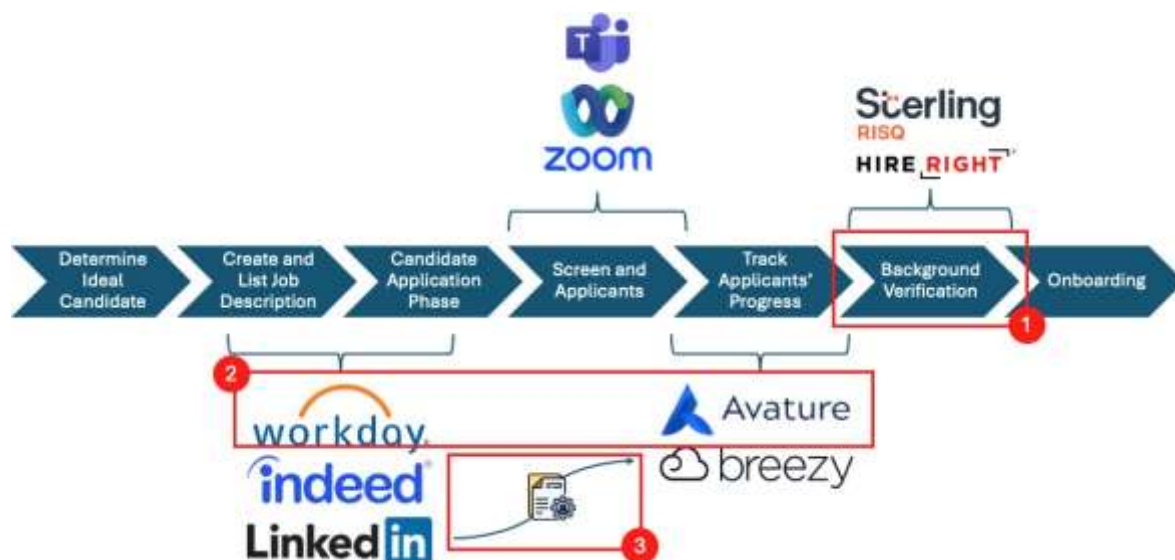


Figure 2.2: Overview of Typical Job Hiring Processes and Potential Issues

2.2.1 Inefficiency in Background Verification Checks (P1)

While background verifications are generally effective, they are not infallible and may contain inaccuracies, in particular false positives, due to outdated data or discrepant data (Indeed, 2024).

The process itself is also highly inefficient. Typically, institutions issue academic certifications to candidates, who must then submit them to prospective employers. Employers, in turn, are then expected to verify these credentials with the issuing institution. However, this verification step is sometimes skipped, creating opportunities for fraud. Additionally, manual screening processes are prone to human error, which can cause legitimate applications to be overlooked.

Strict data privacy regulations further complicate the process by requiring employers to obtain explicit candidate consent before accessing academic records, leading to delays, increased administrative burden and higher operational costs, especially since background checks often occur just before onboarding (Tan-Fernandez, 2024). Clearly, a more objective verification standard is needed.

2.2.2 Inefficiency and Cost of Fragmented Use of Hiring Softwares (P2)

The use of multiple, distinct hiring softwares slows down recruitment as well. Candidates browse job listings on LinkedIn or Indeed, while Applicant Tracking Systems (ATS) like Avature and Breezy track interview progress (St-Jean, 2025). Upon selection, background verification platforms such as Sterling Risk and HireRight step in (Mujibah, 2024).

This fragmented approach results in duplicated data storage, possible manual processing, and increased risks of overlooking critical candidate information. Customizing multiple software solutions further raises costs. Large enterprises spend an estimated US\$125,000 annually on decentralized hiring systems (Mellado, 2025). According to Deloitte, the average cost of hiring a new employee is approximately US\$4,000 (Cision PR Newswire, 2015).

2.2.3 Data Vulnerability in Storage and Transit (P3)

Beyond inefficiency and cost, data security remains a major concern. Is applicant data securely stored and transferred? Are encryption and cybersecurity best practices consistently followed? The reliance on multiple platforms heightens risks such as phishing and insider attacks, endangering both candidates and corporate reputations (Green, 2023).

For instance, in 2024, Amazon confirmed a data breach in which a third-party vendor, MoveIt, exposed sensitive employee information. Such incidents highlight the vulnerabilities of decentralized hiring systems and the need for stricter security measures that today's hiring processes lack.

From the candidate's perspective, the current process is also far from ideal. Applicants often face complex, time-consuming workflows involving repetitive forms, long assessments, and little visibility into the hiring journey.

2.3 Formulating Opportunity Statements

To tackle the identified problems P1 to P3, our team formulated corresponding opportunity statements that would define the scope of our solution and what it aims to achieve.

No.	Opportunity Statement
OS1	Enhancing Efficiency in Background Verification How might we streamline the background verification process to reduce delays and improve reliability while ensuring accuracy and compliance with privacy regulations?
OS2	Integrating Hiring Platforms for a Seamless Process How might we create a more unified hiring system that eliminates redundant data storage, minimizes manual processes, and reduces overall hiring costs?
OS3	Strengthening Data Security in Hiring Systems How might we enhance data security across decentralized hiring platforms to prevent breaches, ensure safe data transfers between candidates and employers, and protect one's personally identifiable information?

Figure 2.2: Overview of Opportunity Statements for this Project

3. Market and Competitor Analysis

3.1 Market Analysis

Building on the opportunity statements stated earlier, we identified consumer wants and needs that align closely with each of them. The table below highlights what students and employers are looking for in the hiring process and the potential for our solution to bridge those gaps.

Opportunity Statement	Consumer Demands	Opportunity to Bridge the Gap
OS1: Enhancing Efficiency in Background Verification	With the increasing prevalence of resume fraud, employers want to hire with confidence, knowing that the applicants' qualifications are authentic. Honest job seekers also want a way to verify their skills and qualifications to gain credibility (Lee & Leu, 2024).	There is strong potential for a system that provides credential verification, building trust and ensuring fairness in the hiring process.
OS2: Integrating Hiring Platforms for a Seamless Process	Instead of utilising multiple softwares, employers are looking for integrated hiring platforms that can reduce costs and speed up the hiring process. Similarly, students seek greater convenience and reduced repetitiveness in form-filling across multiple job application platforms (Hartwell et al., 2020).	This provides an opportunity for an all-in-one integrated platform for both students and employers, which reduces redundancy and allows for a smoother hiring process.
	Employers want to find qualified individuals efficiently without having to manually screen through applications. This is seen in their growing interest in embracing the use of technology to speed up screening and interview processes (Reeracoen Singapore, 2023).	By using AI-powered skills matching to shortlist the most relevant applicants for the job, it can potentially enhance job alignment and efficiency in early-stage screening.
OS3: Strengthening Data Security in Hiring Systems	Both employers and students seek greater data privacy and security during the hiring journey to protect their reputation (Emptor, 2023).	This highlights the need to implement strong authentication and data protection mechanisms to protect users' sensitive information.

3.2 Competitor Analysis

The figure below shows the comparison between the key competitors in the current job application and hiring technology market, based on the features valued the most by employers and job seekers. While they each offer strong features, it is clear that none of the competitors provide a holistic platform that caters to all the consumer demands.

Competitor Analysis				
SmartResume SmartResume is a resume-building platform that allows students to generate resumes that are directly linked to verified credentials . They work with educational institutions to verify academic records.		SeekOut SeekOut is a platform that allows companies to search for suitable candidates through AI-driven skills matching with public profiles across multiple databases, such as LinkedIn and GitHub.		
APPII APPII is a blockchain-based credential verification platform that allows users to create their digital profiles with verified credentials. To ensure data security, they utilise biometric verification for user authentication and store data via blockchain hashing .		Hirex Hirex is an all-in-one integrated platform . It provides a combined user experience for both job seekers and students by covering the end-to-end hiring process within a single platform. It also uses AI-driven candidate matching to enhance hiring efficiency.		
Competitor/ Feature	Credential Verification	All-in-one Integrated Hiring Platform	AI-driven Screening and Matching	Authentication and Data Security
SmartResume	✓			
SeekOut			✓	
APPII	✓			✓
Hirex		✓	✓	

Figure 3.2: Feature Comparison of the Key Competitors

Looking at the gaps identified in the existing platforms, there is a clear need for a solution that addresses the full range of student and employer needs, and that is where our solution, ResumeHub comes into play. The next section outlines the core functionalities of ResumeHub to overcome the limitations of current market offerings.

4. Proposed Solution

To address the opportunity statements that we set out earlier, our group created ResumeHub – An end to end solution and an integrated platform that bridges the hiring gap between employers and students. The following sections will discuss the user features and key technologies of ResumeHub.

4.1 Core User Features

The main functionalities of ResumeHub are summarised in the table below:

Feature	Description
Token-Based/Wallet-Based Authentication	Students log in securely via a blockchain-based wallet and companies log in securely via a token-based authentication system
Resume Builder with Blockchain-Based Credential Verification	Resumes with verified credentials are generated – ResumeHub extracts academic information from OpenCert transcripts uploaded by students, and directly reads the blockchain for Co-Curricular Activities (CCA) records, certifications uploaded by institutions and employment history uploaded by companies.
Resume Customisability	Students can customise their resumes by editing the descriptions, but headers and credentials cannot be edited to ensure authenticity and credibility.
AI-Powered Recommendation System	Based on students' skills and qualifications on their resume, AI algorithms match students with relevant job listings and companies with suitable candidates.
AI-Powered Chatbot	With ResumeHub's integrated chatbot, employers can search for the most relevant applicants based on specific skill requirements using natural language queries.
Dashboard Summary Pages	Once suitable candidates are identified, employers can navigate to a summary dashboard that presents both individual applicant profiles and an overview of the number of applicants matching the queried skills. If needed, employers can also request an AI-generated summary of an applicant's profile, which can be conveniently delivered to their work email for storage or further review. On the applicant's side, the dashboard consolidates all submitted applications, providing a clear and organized view of their application history.
All-in-One Integrated Job Platform	Overall, ResumeHub serves as a convenient one-stop platform that seamlessly integrates the hiring workflow, including resume generation, skills matching, and job applications.

In the user journey diagrams below, we can see some of the functionalities of ResumeHub and how these key functionalities are integrated into the user experience of both students and companies.

4.1.1 Students

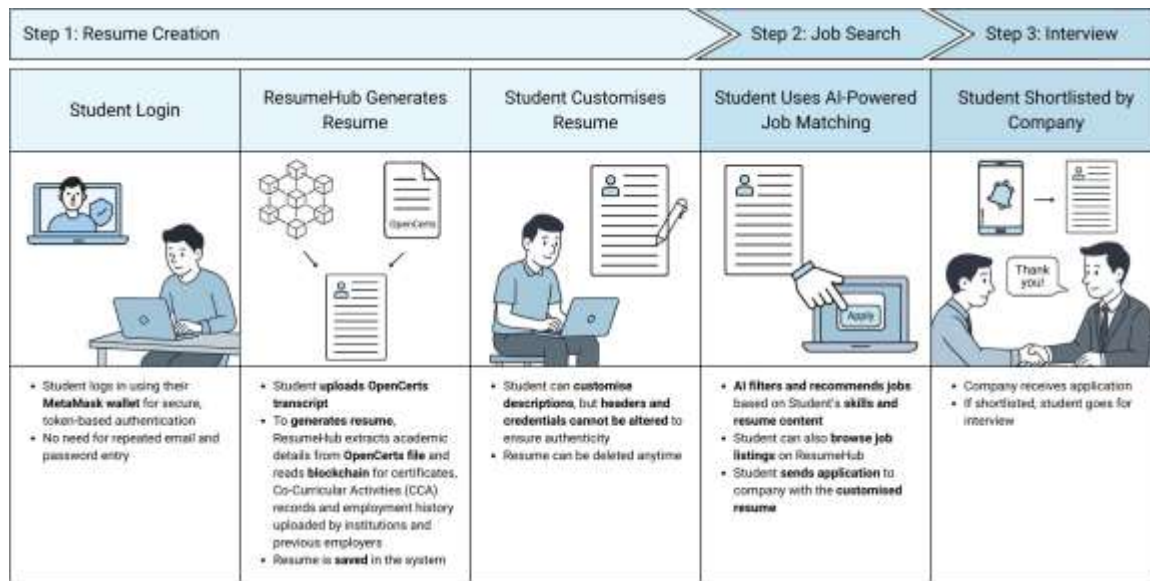


Figure 4.1.1: User Journey for Students

4.1.2 Companies

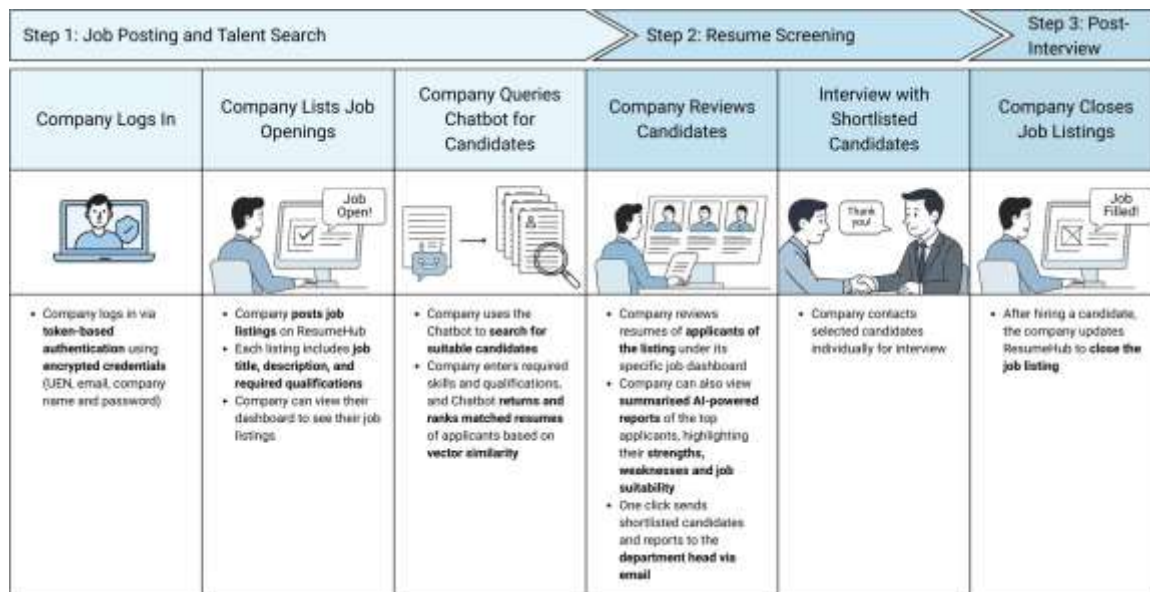


Figure 4.1.2: User Journey for Companies

4.2 Core Technologies

For ResumeHub, each step in the user journey is made possible through the core technologies that we have explored in our course.

4.2.1 Vector Embeddings

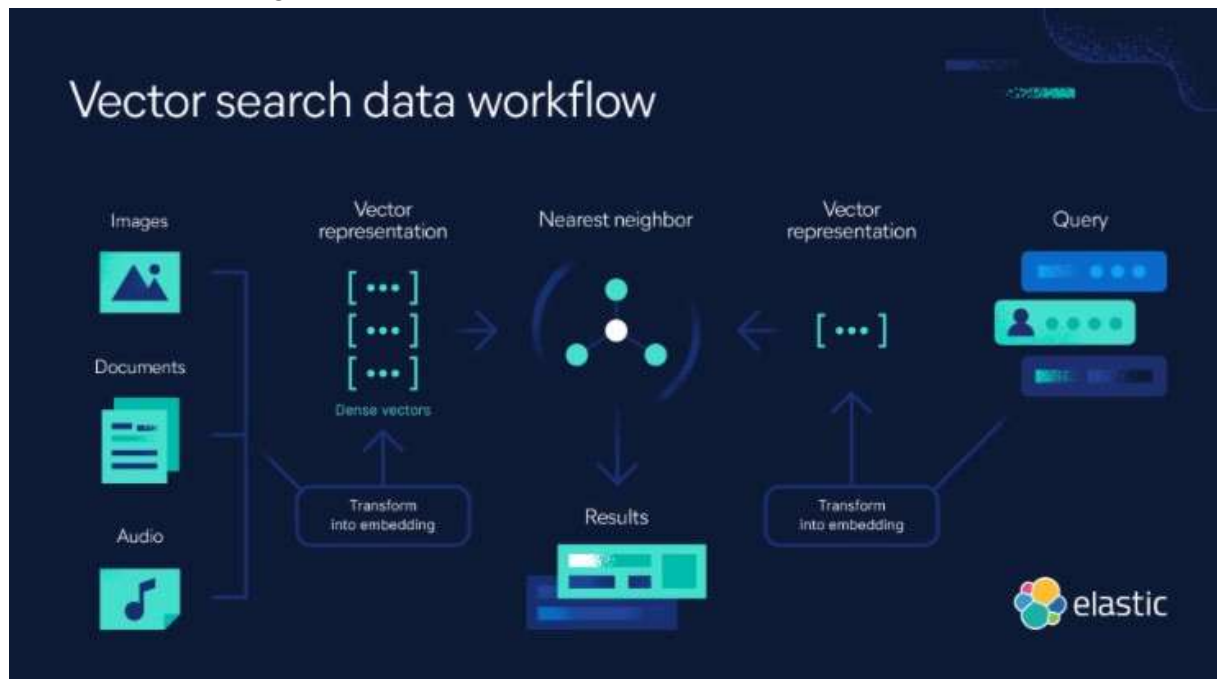


Figure 4.2.1: Illustration of Vector Embedding (Elastic, n.d.)

The first core technology leveraged in ResumeHub is vector embeddings, a method of converting textual and structured data (e.g., words, sentences, and documents) into high-dimensional numerical representations that preserve semantic relationships. In the context of ResumeHub, both word-level and document-level embeddings are utilised to enable efficient semantic search and candidate matching.

For example, upon uploading OpenCerts transcripts, academic details are parsed and transformed into vector embeddings using pre-trained language models. These embeddings are then stored in the vector database, where the dimensionality of each embedding vector corresponds to the underlying model architecture, and the number of vectors stored scales with the quantity and granularity of the academic data extracted.

Also, when a query is initiated, such as searching for candidates with specific qualifications or skills, the system performs a vector similarity search (i.e. using cosine similarity) to identify candidates whose embedded academic profiles exhibit the highest semantic relevance to the query, which allows the appropriate profiles to be pulled to the employer.

4.2.2 GenAI and RAG

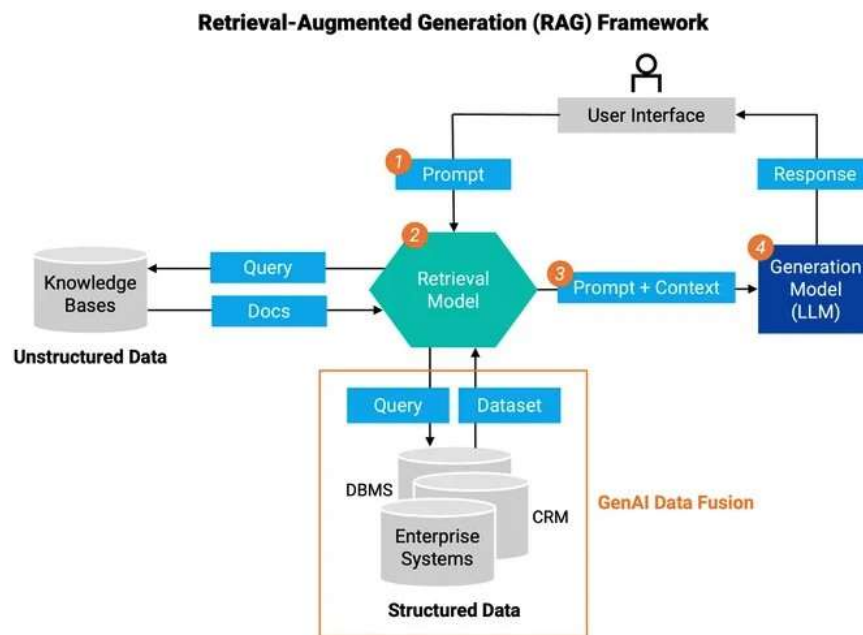


Figure 4.2.2: Illustration of RAG (Erza, 2024)

Another foundational technology employed in ResumeHub is Retrieval-Augmented Generation (RAG), a hybrid architecture that combines traditional information retrieval techniques with the generative capabilities of Large Language Models (LLMs). RAG enhances an LLM's ability to produce contextually accurate and up-to-date responses by dynamically retrieving relevant data from external knowledge bases and injecting it into the prompt context at runtime.

In ResumeHub, the RAG pipeline is primarily exposed through the chatbot interface, which enables companies to query the system for suitable candidates based on specific skill sets or criteria. The workflow proceeds as follows, and can be matched to the numbers above in Figure 4.2.2:

1. **User Prompt (Step 1):** The user initiates a query through natural language input, specifying desired candidate attributes such as skills, qualifications, or experience.
2. **Semantic Retrieval (Step 2):** The query is encoded into a vector representation using the same embedding model explained earlier. This vector is used to perform a semantic similarity search (i.e. using Elasticsearch with cosine similarity) against the embedded corpus of candidate profiles and academic records.
3. **Context Augmentation (Step 3):** The top-k retrieved documents are concatenated and structured into an enriched context window, which is then appended to the original user prompt.
4. **LLM Generation (Step 4):** This augmented prompt is sent to the LLM (i.e. OpenAI's GPT models through an API call), which generates a coherent, contextually grounded response, such as a list of matching candidates or a synthesized resume that aligns with the queried criteria.

By combining dense retrieval with generative modeling, the RAG framework allows ResumeHub to offer highly personalised, semantically relevant outputs without the need to fine-tune the base LLM.

4.2.3 Blockchain and Web3

Thirdly, ResumeHub adopts Web3 and blockchain technologies to move beyond the limitations of traditional Web2 systems and enable secure, user-centric data management. While Web2 relies heavily on centralized infrastructure and data silos controlled by platform owners, Web3 introduces a decentralized model where users retain control over their identity and credentials.

Web3 refers to the next generation of the internet, built on decentralized protocols that prioritize data ownership, transparency, and trustless interactions. At its core is the blockchain, a distributed ledger that records transactions immutably across a peer-to-peer network using consensus mechanisms such as Proof of Stake. This architecture eliminates the need for intermediaries, reduces the risk of data tampering, and enhances user privacy, which links well with our earlier opportunity statements.

In ResumeHub, blockchain is leveraged to enable decentralized identity, verifiable academic and employment credentials, and secure access control. Students can authenticate using a Web3 wallet, and credentials are issued as non-transferable tokens on-chain, ensuring permanent ownership and integrity. However, recognizing the current limitations in user familiarity and enterprise adoption of Web3, ResumeHub strategically integrates Web2 technologies as well, such as traditional authentication, centralized databases (PostgreSQL), and RESTful APIs.

Fully decentralized systems often face adoption and infrastructure challenges, which ResumeHub overcomes by blending Web2 reliability with Web3 security. This hybrid approach allows ResumeHub to be in the Web 2.5 layer and bridge the gap between conventional infrastructure and emerging decentralized technologies. It also ensures backward compatibility with existing institutional systems while introducing the benefits of Web3 in a gradual, user-friendly manner. By doing so, ResumeHub offers a pragmatic and forward-thinking architecture that is both accessible today and scalable for the future.

Next, we will serve to explore more in depth the technical architecture behind ResumeHub in the following section and understand why and how we utilised them.

5. Technical Architecture

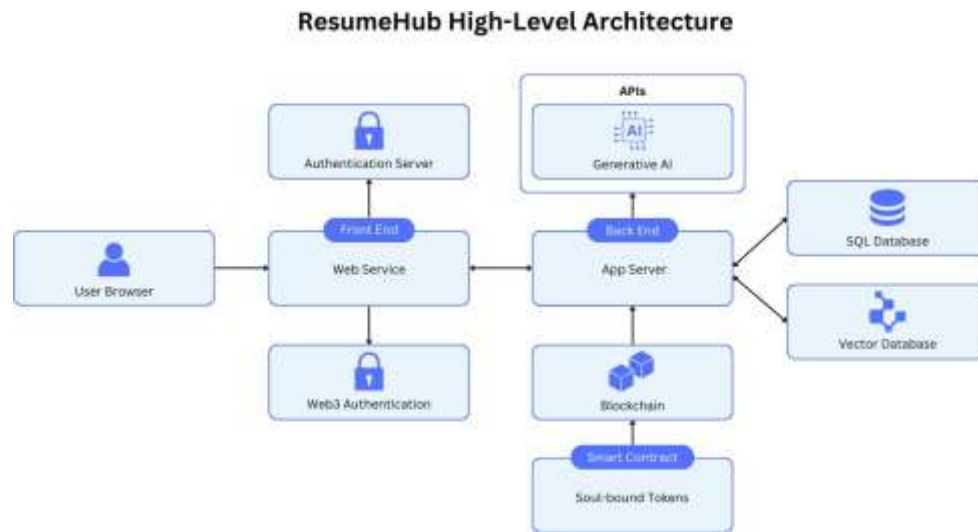


Figure 5.1: High-Level Overview of ResumeHub's Architecture

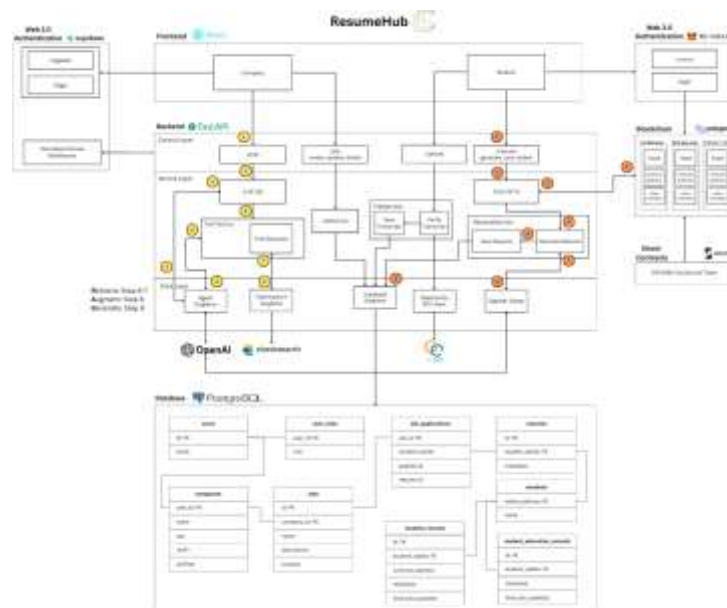


Figure 5.2: Low-Level Overview of Technical Architecture

To support OS2 to build an integrated hiring platform, ResumeHub actually adopts a modular, secure, and scalable technical architecture designed to meet real-world business needs such as system reliability, ease of maintenance, and high performance under scale. Each technology was selected for its scalability, performance, and proven applicability in production environments.

The frontend is developed with React, a component-based JavaScript library that enables rapid UI development and promotes interface consistency. This accelerates development while delivering a

responsive, intuitive user experience. During early stages, we used mock data and local state management to allow us to work and develop on the frontend independently of the backend.

The backend is powered by FastAPI, a high-performance Python framework chosen for its asynchronous capabilities and automatic OpenAPI documentation. We implemented endpoints asynchronously to support concurrent requests, which is particularly important for features like the resume builder and chatbot. FastAPI also provides auto-generated Swagger documentation, streamlining internal testing and integration.

In our prototype, Supabase serves as the backend-as-a-service, handling authentication, credential storage, and file management. It is built on PostgreSQL. This relational database supports complex queries and indexing, making it more efficient and scalable than traditional NoSQL alternatives, particularly for structured data such as academic records and job applications.

Beyond supporting OS2 with this integrated yet modular system, ResumeHub also leverages additional technologies that address OS1 and OS3, as summarised in the table below.

No.	Technology	Purpose	Business Value	Linked Opportunity Statement
1.	Metamask (Web 3.0)	Decentralized login mechanism that uses blockchain-based wallet signatures to authenticate users without relying on centralized identity providers.	Gives users control over their identity and access credentials in a decentralized, verifiable manner.	OS3
2.	Polygon Blockchain	Low-cost, EVM-compatible chain used to mint and store credential metadata as NFTs. Selected for its scalability and low transaction fees.	Enables secure, verifiable credential issuance while minimizing operational costs.	OS1
3.	Soulbound Tokens (SBTs)	Non-transferable NFTs defined under ERC-5484. Used to ensure credentials remain permanently linked to the original recipient.	Prevents falsification or resale of credentials, maintaining the integrity and exclusivity of records.	OS1/OS3
4.	Solidity (Smart Contracts)	Language used to define logic for minting and managing SBT credentials. Contracts enforce ownership, access control, and metadata structure.	Automates credential issuance with secure, verifiable, and auditable logic on-chain.	OS1/OS3
5.	OpenAI API	Processes natural language queries and uses embedding models to transform search inputs into vector form for semantic retrieval.	Enables intelligent, context-aware search and matching through Retrieval-Augmented Generation (RAG).	OS1

6. Value Proposition and Core Principles

In fact, what makes ResumeHub stand out is its strong value proposition and again, business values which guide our design choices.

6.1 Value Proposition

Some of the key considerations on how ResumeHub delivers value to job seekers and employers across key aspects of the hiring process are shown below.

Value	Elaboration
Early Trust Establishment	Unlike traditional hiring methods that conduct background verification after successful interviews, ResumeHub verifies the academic credentials upfront via blockchain. This ensures that only verified candidates will progress through the hiring process, which reduces candidates from spamming applications, allowing employers to receive higher quality applications and better candidates while building mutual trust from the beginning. According to PwC's 2024 report, blockchain-based credential verification reduced hiring risks by 50% within the first year of adoption (Bhandari, 2024).
Safe Credential Authentication	Since all the transcripts are issued as consensual Soulbound Tokens (SBTs), it ensures that all the credentials accessed by employers are verified and tamper-proof. This minimises inefficiencies as employers can choose not to rely on external third-party verification services to verify academic records by using ResumeHub. Furthermore, institutions no longer need to store physical and digital copies of candidates' certifications as they can be accessed from the blockchain, reducing the costs of data storage.
Robust Resume and Job Filtering	Similar to current resume screeners, companies can query the chatbot for resumes and students with relevant skills. The chatbot will fetch and return a list of suitable candidates in the system. By using blockchain as an immutable knowledge base, it prevents the AI model from "hallucinating", reducing errors and increasing the accuracy of skills matching. On top of streamlining the hiring process, AI-powered search reduces unconscious bias as candidates are evaluated solely based on merit during the skill matching stage. This creates a more diverse candidate pool and potentially boosts workplace productivity (FastJobs, 2024).
Integrated User Experience	By consolidating credential authentication, job searching and matching into one integrated platform on ResumeHub, it eliminates the need for multiple software tools. This reduces hiring costs and streamlines the job application process for both candidates and employers.
Privacy in Transparency	ResumeHub provides transparency in credential verification while ensuring data privacy through decentralised ownership of credentials. Candidates still have control over access to their personal credentials, preventing unnecessary data exposure. This minimises the risk of data breaches and leaks for both employers and candidates, which can be costly and damaging (Singh, 2024).

6.2 Business Value

Beyond solving the pain points in the hiring process, ResumeHub also provides long-term business values.

In fact, ResumeHub aligns with the Singapore government's Career & Skills Passport (CSP) initiative, which is scheduled to roll out in Q2-Q3 2025. CSP aims to create a digital record of one's skills, employment records, academic qualifications and certificates from verified Government sources. Individuals can access their profile through SingPass and share it with online job portals (SkillsFuture Singapore, n.d.). ResumeHub can complement the CSP initiative by providing a platform to close the loop between the CSP initiative and the private job market, streamlining the hiring pipeline and reducing risk of using third-party platforms. By integrating SingPass authentication to securely verify both individuals (via NRIC) and companies (via UEN), users can directly apply for jobs on ResumeHub using their CSP. This can be supported by a system that integrates with the CSP using trusted APIs and blockchain-based standards.

By linking individuals' CSP directly to employers, ResumeHub allows for a self-sustaining ecosystem where verified skills are continuously updated and utilised. As CSP gets widely adopted and NFT-based credentials become increasingly prioritised, ResumeHub has the potential to become the first platform to fully integrate the NFT wallet.

ResumeHub also has strong economic and monetisation benefits. As verified qualifications become increasingly valued, individuals will be motivated to obtain NFT versions of their credentials. This creates opportunities for academic institutions and certificate providers to generate additional revenue by offering these certification services. By minting these NFTs and charging a small fee, it ensures a sustainable business model, especially for SMEs. From companies' perspective, NFT-backed resumes reduce the risk of resume fraud and ensure higher quality applications from qualified individuals. This significantly lowers hiring costs and reduces the time spent on manual screening.

Lastly, ResumeHub is built on the OpenCerts standard, which is recognised both in Singapore and globally. This allows its infrastructure to be expanded on a global scale in the future. The blockchain-based credentials can be easily recognised by both local and international institutions, allowing for greater cross-border job mobility and credential verification. ResumeHub is also technically scalable as the NFT-based credentials do not require physical storage. Therefore, the platform can support large volumes of users without compromising efficiency or data security.

7. Feasibility Analysis

Yet, while ResumeHub is designed to be an ideal solution, we acknowledge potential challenges that may affect its feasibility. This section outlines key considerations and our strategies to address them.

7.1 Financial

The global recruitment market, valued at USD 757.56 billion in 2024, is projected to grow to USD 2,297.45 billion by 2032, reflecting a compound annual growth rate (CAGR) of 13.1% (Business Research Insights, 2025). Simultaneously, the blockchain industry is expected to reach USD 746.41 billion by 2032, growing at a CAGR of 49.7% (Fortune Business Insights, 2025), while the Web3 market is forecasted to expand from USD 0.4 billion in 2023 to USD 5.5 billion by 2030, at a CAGR of 44.9% (MarketsandMarkets, 2023).

These figures point to a rapidly growing intersection between recruitment and decentralized technologies, one that ResumeHub is strategically positioned to capitalize on. For small and medium-sized enterprises (SMEs), adopting ResumeHub offers a financially viable solution to streamline hiring processes. Traditional recruitment software can cost anywhere from \$15 to \$75 per user per month, with basic plans starting around \$200–\$300 monthly (Bartram, 2024). These recurring costs can be difficult for smaller businesses to sustain, especially when hiring needs fluctuate. ResumeHub, on the other hand, leverages lightweight infrastructure and requires only a few API calls per verification, significantly reducing overhead.

We aim to offer a flat subscription starting at \$49 per month for up to five active job listings, with no per user charges or hidden fees. This pricing structure is intentionally simple and transparent, removing the complexity and high overhead costs that often deter smaller businesses from adopting recruitment softwares. Our proposed pricing scheme would also encompass the fees needed to mint the NFTs and mentioned earlier. By significantly undercutting traditional pricing models while still delivering automated credential verification, streamlined workflows, and secure candidate profiles, ResumeHub directly addresses the pain points SMEs face. This value proposition positions ResumeHub as a compelling solution that we believe will resonate strongly with SMEs looking for efficiency, trust, and affordability in their hiring processes.

7.2 Technical

One key technical challenge we identified is the performance of high-dimensional vector searches, particularly as the volume of resumes and academic records increases. Semantic search queries can become computationally intensive and degrade response times without proper indexing and optimization. To address this, we have strategically chosen Elasticsearch as our retrieval engine, as it supports approximate nearest neighbour (ANN) using dense vector fields, which allows us to perform fast and scalable similarity searches. While our current implementation is tailored for smaller datasets, the use of Elasticsearch allows for greater scalability without compromising performance as data volume could potentially increase in the future as more users adopt ResumeHub.

Another challenge lies in ensuring data consistency across both Web2 and Web3 components. Integrating decentralized technologies such as blockchain introduces potential issues with latency and synchronization, particularly when managing credential issuance and verification. In the current prototype, we simulate the blockchain integration and design the system to accommodate confirmation and state tracking mechanisms in future releases. This ensures the architecture is forward-compatible while maintaining system reliability in the present deployment.

7.3 Regulatory

Since ResumeHub is planning to integrate with government platforms such as Singpass, this positions us well to meet the requirements of Singapore's Personal Data Protection Act (PDPA). Such integration supports strong user authentication, explicit consent collection, and purpose-bound data access, which are the core principles of the PDPA (PDPC, 2022).

In line with this, all access to government-linked data will be gated by user authorization, and used strictly for credential verification within the platform. Sensitive data will not be stored on-chain. Instead, only hashed references or anonymized metadata will be recorded, while full records remain encrypted in secure off-chain storage.

We also acknowledge potential intellectual property concerns when using institutional formats or content. To address this, we have plans to establish contractual data-sharing agreements with partner institutions to ensure proper authorization and ethical use of shared materials.

8. Conclusion

All in all, while we recognise that the adoption of Web3 tools like the MetaMask wallet could be a potential hurdle, particularly for users unfamiliar with blockchain or those who find the setup process cumbersome. In fact, we do not expect every user to immediately adopt this solution, but rather our goal is to encourage the majority to transition smoothly by actively promoting the benefits of decentralised identity, data ownership, and verifiable credentials. Through user education, seamless onboarding, and continued exposure to the value of Web3, we aim to drive gradual adoption over time.

Additionally, at this stage, features such as employment history and transcript uploads are simulated using mock data. This design choice allows us to demonstrate functionality while keeping the system adaptable. Once ResumeHub achieves stable adoption, these features, including the aforementioned CSP initiative and Singpass authentication system, can be fully integrated with institutional and/or company databases, enabling seamless data population and expanding the platform's real-world applicability with ease.

Overall, ResumeHub represents a forward-thinking project and solution that merges AI, Web3, and modern Web 2 technologies to solve real-world challenges in resume building and candidate verification, whilst leveraging from what we have learnt from this module. By balancing innovation with accessibility, and automation with trust, ResumeHub is well-positioned to make a meaningful impact in how students and job seekers present themselves and how employers find talent in today's increasingly digital age.

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